

Cell culture Experiments

Lab protocol (Version 2.4)

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Last modification 21st June 2017

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I. Cell culture procedures

1. Before starting cell culture : General considerations

- Before starting any experiment, wash the hood with 70% ethanol.
- Never use gloves that were used in the main lab. In case you have to leave the hood, always spray the gloves with 70% ethanol before starting any manipulation.
- Clean with ethanol everything that goes in and out of the hood: box of tips, culture flask, bottle of medium, trypsin tube, etc..For hemacytometer: It must be rigorously cleaned with ethanol between uses.
- Manipulate more possible depth of the hood
- Don't cough or sneeze facing the hood. Also, don't avoid or talking facing an incubator when it is opened.
- Never move the arms or equipment over a bottle or a plate open.
- After your manipulation don't forget to :
 - Wash the surface of the hood with the « Vim ».
 - Rinse thoroughly with water.
 - Rinse thoroughly with ethanol and dry.

2. C2C12 cell line

- The cell line C2C12 is an immortal line of mouse skeletal myoblasts originally derived from satellite cells from the thigh muscle of a two month old female C3H mouse donor 70h after a crush injury (*Yaffe and Saxel., 1977, Karyotyping., 2011*). From the C2s the immortal subline C2C12 was selected (*Blau et al., 1985*).
- C2C12 myoblasts are often used *in vitro* to study the muscle development and differentiation because they have all the characteristics needed for the study of myogenesis: differentiates rapidly, forming contractile skeletal myotubes and producing characteristic muscle proteins and the androgen receptor (AR) (*Bains et al., 1984, Bennett et al., 1997*).
- The ability of C2C12 cells to differentiate *in vitro* means that these cells represent a valuable source of material for a wide variety of studies examining skeletal myoblasts and myotubes in muscle biology. In addition to studies directly examining myogenesis and gene regulation (*Iyer et al., 2006, Ono et al., 2006*), applications of C2C12 cells and differentiated skeletal myotubes include studies examining insulin signaling (*Kumar et al., 2002*), metabolic activity (*Weber et al., 2002*) electrophysiology (*Fioretti et al., 2005*), oxidative stress (*Rohrbach et al., 2006*), cell migration (*Goetsch et al., 2005*) signal transduction (*Ko WC et al., 2006*), hypertrophy (*Cross-Doersen et al., 2003*) and tissue replacement (*Laino et al., 2006*).

3. C2C12 cell culture :

- C2C12 is a fairly quickly growing cell line (**doubling time is approximately 12h**); **we will need to passage them every 1-2 days.**

4. Medium and reagents preparation under clean bench :

- **Fetal Bovine Serum (FBS)**
 - Defreeze 500 ml of a FBS bottle (Hyclone, stored at -80°C) in 37°C water bath.
 - Prepare 10 x 50 ml Falcon tubes (FBS, 1/10-10/10, date) and 50 ml pipette.
 - Aliquot 50 ml with a 50 ml pipette to 50 ml Falcon tubes (total 10 tubes).
 - Store at -20°C.
- **Penicillin-Streptomycin solution**
 - Defreeze 100 ml of a 100X Penicillin-Streptomycin bottle (Sigma Cat# P0781, Stored at -20°C) in 37°C water bath.
 - Prepare 10 x 15 ml Falcon tubes (100X PS, 1/10, date) and 50 ml Eppendorf Conbitip Plus Tip.
 - Aliquot 10 ml with a 50 ml Eppendorf Conbitip Plus Tip (Set channel at 5) to 15 ml Falcon tubes (total 21 tubes).
 - Store at -20°C.
- **10% FBS Dulbecco's Modification of Eagle's Medium (10% FBS-DMEM, myoblast growth medium) for 100 ml :**
 - Incubate 89 ml of DMEM, 10 ml of FBS (Stored at -20°C in a 50 ml Falcon tube), and 1 ml of 100X Penicillin-Streptomycin Solution (Store at -20°C in a 15 ml Falcon tube) in 37°C water bath.
 - Add 10 ml of FBS and 1 ml of 100X Penicillin-Streptomycin Solution into 89 ml of DMEM (10%FBS-DMEM, date).
- **Medium for freeze:** 90% FBS + 10% Dimethyl sulfoxide (DMSO), for 10 ml
 - Add 9 ml of FBS + 1 ml of DMSO (Sigma Cat# D2650)

*Cells can be frozen for several months at -80°C or for years in liquid N₂.

**Using the Cell Banker (ZENOAQ Cat# BLC-1), cells can be frozen for 2 years at -80°C (Freezing medium, date).
- **1X Phosphate buffer solution (PBS)**
 - Prepare each 1000 ml bottle and 1000 ml cylinder (500 ml can be used), and 2 bottles of 500 ml bottle with filter 0.22 µm (1X PBS, 1/2-2/2, date).
 - Add 100 ml of 10X PBS (Invitrogen Cat# 70011-044) + 900 ml of DW in a 1000 ml bottle.
 - Filter with 2 bottles of 500 ml bottle with filter.
- **0.05% Trypsin-Ethylenediaminetetraacetic acid (EDTA) solution**
 - Prepare 1000 ml bottle and 1000 ml cylinder (500 ml can be used).
 - Defreeze 100 ml of a 0.5% Trypsin-EDTA bottle (Sigma Cat# T4174, store at -20°C) in 37°C water bath.
 - Prepare 7 x 15 ml Falcon tubes (0.05% Trypsin-EDTA, 1/7-7/7, date) and 50 ml pipette.
 - Add 100 ml of 0.5% Trypsin-EDTA (Sigma Cat# T4174, store at -20°C) + 900 ml of 1X PBS in a 1000 ml bottle.
 - Aliquot 13 ml with a 50 ml pipette to 15 ml Falcon tubes (total 7 tubes).
- **Diluted Vim**
 - Take 500 ml of DW.
 - Add 1 plug of Vim solution.

5. Maintenance of CO₂ incubator:

- **Change water each 2 wks.**
 - **Trash water in the bat and clean the bat with 70% ethanol.**
 - **Add 1 l of DW from 1 l bottle into the bat.**
 - **Add 10 ml Lysol with a 15 ml Falcon tube into 1 l of DW.**
- **Check CO₂ concentration every wk.**
 - **Stand up CO₂ meter straight.**
 - **Connect a black rubber pump to sample plug of the incubator.**
 - **Squeeze the pump 3X.**
 - **Incline CO₂ meter upside down at 45 degree and then stand up CO₂ meter straight.**
 - **Push the top of the CO₂ meter and check if the level of red liquid is at 0.**
 - **Connect pump and the top of the CO₂ meter** (Push the top of the CO₂ meter down with the pump connection part).
 - **Squeeze the pump 18X while holding down the connection part.**
 - **Check the level of red liquid** (CO₂ level should be in the range of 4.5 to 5.5).
 - **Remove the connection.**
 - **Repeat from “inclining the CO₂ meter”.**
 - **If CO₂ level is not in a range, ask for repare.**

6. Preparation of clean bench before use

- **Wet a paper towel with water and put Vim.**
- **Wipe surface of clean bench with Vim side of the paper towel.**
- **Wipe surface of clean bench with water side of the paper towel.**
- **Spray 70% ethanol to the surface of clean bench.**
- **Wipe all materials which will use in the clean bench with 70% ethanol.**

7. Washing glass wares:

- **After use, rinse with hot water.**
- **Place glass wares in a washing machine.**
- **Put 20 ml Glass-Klenz (STERIS Cat# 1114-08) in the washing machine.**
- **Start the machine.**
- **Rinse with DW 20X.**
- **Autoclave.**

8. C2C12 cell culture thaw :

- It is important to avoid excessive alkalinity of the medium during recovery of the cells.
- Place DMEM and FBS in 37°C water bath for 30 min.
- Two culture vessels containing the growth medium be placed into the incubator for at least 15 min to allow the medium to reach its normal pH (7.0 to 7.6).
- Make 5 ml of warm 10% FBS-DMEM in a 15 ml flacon tube.
- Take out cells from liquid N₂.
- Under the hood, **unscrew the cap and then screw the tube** (This can leave evacuate the stored pressure in the tube).
- **Quickly place in the 37°C water bath** (disposed to the half of the tube in the bath at 37 °C, to reduce the possibility of contamination, keep the O-ring and cap out of the water). **Shake the tube quickly in the bath.**
- Under the hood, immediately transfer all the contents of the cryovial tube (1ml) to the 15 ml flacon tube contains 5 ml of warm medium.
- Centrifuge at 500 G for 5 min.
- Aspirate the medium.
- Resuspend the pellet with approximately 2 ml of warm 10% FBS-DMEM and transferred into two flasks T-75 containing each 10 ml of warm 10% FBS-DMEM.
- Incubate cells at 37°C (5% CO₂).
- Change medium after all cells have adhered (After 48 h).

Note: Thawing should be rapid (Approximately 2 min).

9. C2C12 cell culture passage :

- Put medium and trypsin-EDTA in 37°C water bath for 30 min.
- Prepare flasks and 50 ml Falcon tubes as needed (name of cells, number of passage, date).
- Check the amount of 2.5, 5 and 10 ml pipettes and Pasteur pipettes.
- Observe cells under an inverted microscope to evaluate the confluency.

Note: It is important to not let the cells become fully confluent because they can begin to fuse and partially differentiate upon cell-cell contact.

- To passage the cells: aspirate the culture medium in the flask.
- Rinse twice with PBS.
- Subculture the cells using standard trypsinization techniques : add 3 ml of 0.05% trypsin-EDTA solution. Trash the pipette in the autoclaving bag in the clean bench.
- Incubate at 37°C for 2-10 min (not more than 10 min).
- Observe cells under an inverted microscope until cell aspect changes to round (Usually within 60-90 sec).
- To avoid clumping do not agitate the cells by hitting or shaking the flask while waiting for the cells to detach. Cells that are difficult to detach may be placed at 37°C to facilitate dispersal.
- Add 9 ml of new myoblast growth medium (3 times the volume of the enzyme) in the flask. Mix by pipetting with the 10 ml pipette.
- Collect cells by gently pipetting.
- Transfer the medium containing cells into a 15 ml Flacon tube.
- Take a small volume of cells (Usually we take $10\ \mu\text{l} \times 2 = 20\ \mu\text{l}$) and count cells using a hemocytometer (Step I.10).
- Centrifuge for 5 min at 500G.
- Standup a new flask and add 15 ml (10-20 ml) of new medium into the flask.
- Transfer 2 ml of medium containing cells into the flask containing new medium. (In order to have 1×10^6 cells/mel means 10^6 viable cells/75 cm²).
- Incubate at 37°C.
- If cells have been growing well for a few days but are not yet confluent, then they will require media changing to replenish nutrients and keep correct pH. If there are a lot of cells in suspension (attached cell lines) or the media is staring to go orange rather than pinky orange then media change them as soon as possible. To change the medium, remove the medium and add fresh media :
 - Put medium in 37°C water bath for 30 min.
 - Prepare Pasteur and 10 ml pipettes.
 - Aspirate the medium.
 - Add 15 ml (10-20 ml) of new medium.

Note:

- Mycoblasma test must be done: using Mycoplasma detection kit conventional PCR (VenorGem) (See Appendix 1).

- **If the cells are still attached**, aseptically remove all but 5 to 10 ml of the shipping medium. The shipping medium can be saved for reuse. Incubate the cells at 37°C in a 5% CO₂ in air atmosphere until they are ready to be subcultured.
- **If the cells are not attached**, aseptically remove the entire contents of the flask and centrifuge at 125 xg for 5 to 10 min. Remove shipping medium and save. Resuspend the pelleted cells in 15 ml of this medium and add to 75 cm² flask. Incubate at 37°C in a 5% CO₂ in air atmosphere until cells are ready to be subcultured.
- **Medium renewal**: Every 2 or 3 days (ATCC recommendation).

10. Viability Test :

Trypan Blue is a vital dye. The reactivity of trypan blue is based on the fact that the chromophore is negatively charged and does not interact with the cell unless the membrane is damaged. Therefore, all the cells which exclude the dye are viable.

Note: Trypan Blue has a greater affinity for serum proteins than for cellular protein. If the background is too dark, cells should be pelleted and resuspended in protein-free medium or salt solution prior to counting.

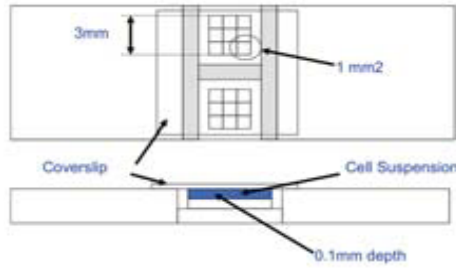
Staining facilitates the visualization of cell morphology.

- **Reagents preparation**
 - **0.4% Trypan Blue solution** (Sigma Cat# T8154)
 - **1X PBS** (Step I.4)
- **Prepare a cell suspension in PBS.**
- **Transfer 0.5 ml of 0.4% Trypan Blue solution** (Sigma Cat# T8154) (w/v) **to a test tube.**
- **Add 0.3 ml of PBS and 0.2 ml of the cell suspension (dilution factor = 5) and mix thoroughly.**
- **Allow to stand for 5 to 15 min at RT.**

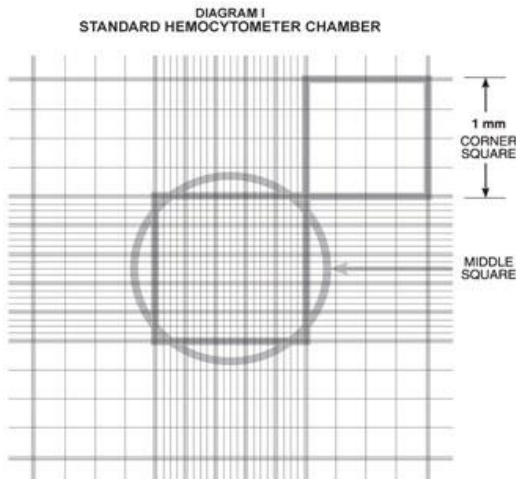
Note: If cells are exposed to Trypan Blue for extended periods of time, viable cells, as well as non-viable cells, may begin to take up dye.

- With the cover-slip in place, **transfer 20 µl of the mixture cell suspension- trypan blue to both chambers of the hemocytometer** (10 µl in each chamber).
- **Carefully touch the edge of the cover-slip with the pipette tip and allow each chamber to fill by capillary action.** Do not overfill or underfill the chambers.
- **Incubate the hemocytometer and cells for 1 – 2 min at RT.** For longer incubations, please use a humid chamber. (Incubations exceeding 30 min may cause decreased cell viability due to Trypan toxicity).
- **Load a hemocytometer and examine it under a microscope at low magnification.**
- **Starting with chamber 1 of the hemocytometer, count all the cells in the 1 mm center square and four 1 mm corner squares** (see **pictures** below). Non-viable cells will stain blue. Keep a separate count of viable and non-viable cells.

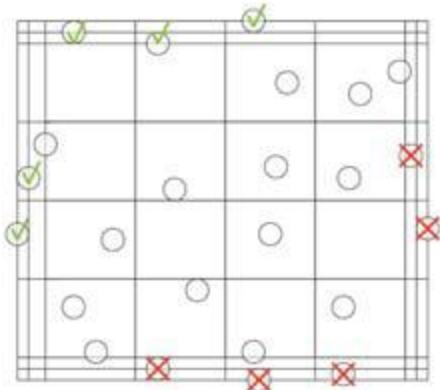
Note: Count cells on top and left touching middle line of the perimeter of each square. Do not count cells touching the middle line at bottom and right sides.



A haemocytometer being filled with cell suspension.



Dimensions of a haemocytometer, the circle in indicates the approximate area covered at 100× microscope magnification (10× ocular and 10× objective). Include cells on top and left touching middle line (O). Do not count cells touching middle line at bottom and right (Φ). Count 4 corner squares and middle square in both chambers (one chamber represented here).



Guide to which cells to count on grid i.e. include cells in middle of grid and cells that overlap edge on two sides (ticked), exclude cells that overlap edge on other two sides (crossed).

- **Repeat this procedure for chamber 2.**

Note:

- If greater than 10% of the cells appear clustered, repeat entire procedure making sure the cells are dispersed by vigorous pipetting in the original cell suspension as well as the Trypan Blue-cell suspension mixture. If less than 200 or greater than 500 cells (i.e., 20–50 cells/square) are observed in the 10 squares, repeat the procedure adjusting to an appropriate dilution factor.
- Cell viability should be at least 95% for healthy log-phase cultures (count the number of all the cells on 4 corners in each chamber).

- **Calculations**

- **Cell Counts** – Each square of the hemacytometer, with cover-slip in place, represents a total volume of 0.1 mm^3 or 10^{-4} cm^3 . Since 1 cm^3 is equivalent to approximately 1 ml, the subsequent cell concentration per ml (and the total number of cells) will be determined using the following calculations:
- **Cells/ml** = (average count of 2 chambers/4 corners) 10^4
- **Cells Per ml** = the average count per square $\times$ dilution factor $\times 10^4$ (count 10 squares).
- **Total Cells** = cells per ml $\times$ the original volume of fluid from which cell sample was removed.
- **Cell Viability (%)** = total viable cells (unstained) $\div$ total cells (stained and unstained) $\times 100$.

11. Cleaning of clean bench after use:

- **Put aseprite in all dispensable tubes and flasks which contain cells. Then trash as an usual trash.**
- **Aspirate aseprite, then 70% ethanol. Trash the liquid in the flask connected to aspirator to sink.**
- **Wet a paper towel with water and put Vim.**
- **Wipe surface of clean bench with Vim side of the paper towel.**
- **Wipe surface of clean bench with water side of the paper towel.**
- **Spray 70% ethanol to the surface of clean bench.**
- **Wipe all materials which have been used in the clean bench with 70% ethanol.**

12. Storage of cells

- **Can store the leftover cells from the Step I.9**
- **Transfer 4 ml of the leftover into a 15 ml Falcon tube.**
- **Centrifuge at 500 G for 5 min and remove supernatant carefully by aspiration.**
- **Add 1 ml of Cell Banker (ZENOAQ Cat# BLC-1) for 2.0×10^6 cells.**
- **Transfer 1 ml of the medium containing cells to a 1.0 ml Nunc cryo tube vial.**
- **Store the tube at -80°C (Can be stored for 2 yrs).**

If using medium for freeze, place the tube at -80°C for 24 h – 1 wk (Can be stored for 2-3 months), then place the tube in liquid N_2 (Can be stored for 10 years).

II. Effects of SPARC on C2C12 cells differentiation, ECM remodeling and mitochondrial functions

1. Number of cells needed

- We want to study if induction or inhibition of SPARC will affect the extracellular matrix (ECM) and mitochondrial function. Thus, we will measure C2C12 cells differentiation (morphology) and study the expression of 6 proteins by western blot: fibronectin (Fn), Collagen (Col) type I, Col III and Col IV, peroxisome proliferator-activated receptor gamma coactivator 1 alpha (PGC1a) and nuclear respiratory factor 1 (NRF1).
- For each well (protein) in western blot we need 30 µg of proteins so in total: **30 µg × 6 proteins = 180 µg.**
- A 'typical' mammalian cell, for example a liver hepatocyte, is thought to contain 10 000–20 000 different proteins, about 8×10^9 individual molecules in all, representing approximately **0.5 ng of protein** or 18–20% of the total cell weight (*Alberts et al.*, 1994, *Lodish et al.*, 2000) so :
 - 1 cell $\longrightarrow$ 0.5 ng of protein = 5×10^{-4} µg
 - X cell $\longrightarrow$ 180 µg of proteins.
 - We need to plate in each well: **number of cells = 3.6×10^5 cells/well.**

	Surface Area (mm ²)	Seeding Density	Cells at Confluency ¹	Versene (ml of 0.53 mM EDTA)	Trypsin (ml of 0.05% trypsin, 0.53 mM EDTA)	Growth Medium (ml)
Dishes						
35 mm	962	0.3×10^6	1.2×10^6	1	1	2
60 mm	2,827	0.8×10^6	3.2×10^6	3	2	3
100 mm	7,854	2.2×10^6	8.8×10^6	5	3	10
150 mm	17,671	5.0×10^6	20.0×10^6	10	8	20
Cluster Plates						
6-well	962	0.3×10^6	1.2×10^6	2	2	3–5
12-well	401	0.1×10^6	0.4×10^6	1	1	1–2
24-well	200	0.05×10^6	0.2×10^6	0.5	0.5	0.5–1.0
Flasks						
T-25	2,500	0.7×10^6	2.8×10^6	3	3	3–5
T-75	7,500	2.1×10^6	8.4×10^6	5	5	8–15
T-160	16,000	4.6×10^6	18.4×10^6	10	10	15–30

http://www.invitrogen.com/etc/medialib/en/filelibrary/pdf/Par.4786.File.dat/Useful_Numbers_Y14472_Useful_Nmbrs.pdf

- **12-well plates can be used because we can plate 4×10^5 cells at confluency/well.**

2. Differentiation treatment :

- Cells will be washed with PBS, and the culture medium will be replaced with differentiation medium.
- Differentiate for 24 hours to 7 days **by rinsing fully confluent cells once with PBS and adding low-serum differentiation medium.**
- Feed with fresh differentiation medium every **24 hours up to the 72h** timepoint and after that, **every 24 hours** (as these cells differentiate, they begin to deplete and acidify the medium more quickly)
- **Myotube formation is enhanced when the differentiation medium is supplemented with 10% horse serum (HS) instead of FBS.**

3. Differentiation medium and reagents preparation:

- **Horse serum (HS)**
 - Defreeze 500 ml of a HS bottle in 37°C water bath.
 - Prepare 21 x 15 ml Falcon tubes (HS, 1/10-10/10, date) and 50 ml Eppendorf Conbitip Plus Tip.
 - Aliquot 10 ml with a 50 ml Eppendorf Conbitip Plus Tip (Set channel at 5) to 15 ml Falcon tubes (total 21 tubes).
 - Store at -20°C.
- **2% HS-DMEM: For 100 ml**
 - Incubate 97 ml of DMEM, 2 ml of HS and 1 ml of 100X Penicillin-Streptomycin Solution (Store at -20°C in a 15 ml Falcon tube) in 37 ° C water baths.
 - Add 2 ml of HS and 1 ml of 100X Penicillin-Streptomycin Solution into 97 ml of DMEM (2 % HS-DMEM, date).
- **10% HS-DMEM: (ATCC recommendation)**
 - Incubate 445 ml of DMEM, 50 ml of HS and 5 ml of 100X Penicillin-Streptomycin Solution (Store at -20°C in a 15 ml Falcon tube) in 37 ° C water baths.
 - Add 10 ml of horse serum and 5 ml of 100X Penicillin-Streptomycin Solution into 445 ml of DMEM (10 % HS-DMEM, date).
- **10%FBS-DMEM** (Step I.4)
- **1X PBS** (Step I.4)
- **Methanol**
- **Hematoxylin solution according to Delafield** (Sigma Cat#03971).
- **Glycerol in PBS (v/v) :**
 - Add 5 ml of glycerol in tube.
 - Add 5 ml of 1X PBS.
 - Vortex gently.
 - Make appropriate label (50% Glycerol in PBS (v/v), date).

4. General procedure for differentiation

- **Prepare differentiation medium (2% HS-DMEM).**
- **Aspirate growth medium**, when cells reach about 50-80% confluence,
- **Switch cells to differentiation medium.**
- **Incubate at 37°C and 5% CO₂.**
- Visualization under light microscopy should reveal myotube formation after **24 h** in differentiation medium.
- **Differentiation medium must be changed each 2 days.**

5. Doubling time of C2C12 and myoblast fusion

In order to confirm the doubling time of our cell line C2C12, its attachment, confluency and myoblast fusion, we will do this experiment.

Appropriate seeding density: Start with an optimal seeding density is also very important for adherent cells.

Fast attachment of adherent cells: Preheating of the medium and the use of large volume flask containing already CO₂ in the medium were recommender in order that the medium will quickly reach its optimum pH, this step can cause a faster attachment.

Fusion: After staining with haematoxylin, the nuclei present in myoblasts and in myotubes were counted.

$$\text{Fusion (\%)} = \frac{\text{Number of nuclei present in myotubes}}{\text{Total number of nuclei (myoblasts + myotubes)}} \times 100.$$

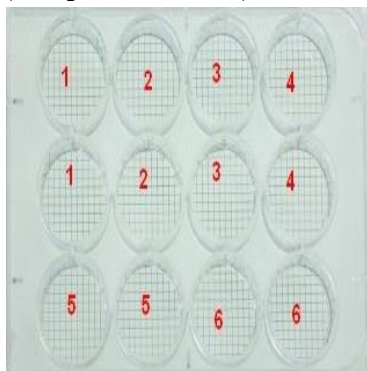
5.1 Reagents preparation

- **10% FBS-DMEM** (Step I.4)
- **1X PBS** (Step I.4)
- **0.05% trypsin-EDTA solution** (Step I.4)
- **2% HS-DMEM** (Step II.3)
- **Methanol**
- **Hematoxylin solution according to Delafield** (Sigma Cat#03971).
- **Glycerol in PBS (v/v)** (Step II.3)

5.2 Procedure of doubling time of C2C12 and myoblast fusion :

[Day 1]: Seed cells

- **Trypsinize cells.**
- **Plate different number of cells** (in growth medium: 10% FBS-DMEM) **in 3X 12-well plates.** (See picture below).



Well	Cells
1	4×10^3
2	4×10^4
3	1×10^5
4	2×10^5
5	3×10^5
6	4×10^5

[Day 2]: Add differentiation medium to plate n°2 and plate n°3

- **After 24h** (3h or 4h for its attachment + 12h doubling time), **Observe cells under an inverted microscope** to evaluate the confluency.
- **For plate n°1 :**
- **Put medium and trypsin-EDTA in 37°C water bath for 30 min.**
- **Aspirate the culture medium in the plate.**
- **Rinse twice with PBS.**
- **Add 0.5 ml of 0.05% trypsin-EDTA solution.** Trash the pipette in the autoclaving bag in the clean bench.
- **Incubate at 37°C for 2-10 min** (not more than 10 min).
- **Observe cells under an inverted microscope until cell aspect changes to round** (usually within 60-90 sec).
- **To avoid clumping do not agitate the cells by hitting or shaking the flask while waiting for the cells to detach.** Cells that are difficult to detach may be placed at 37°C to facilitate dispersal.

- Add 1.5 ml of new myoblast growth medium (3 times the volume of the enzyme) in each well of the plate. Mix by pipetting with the 10 ml pipette.
- Collect cells by gently pipetting.
- Take a small volume of cells (usually we take $10\ \mu\text{l} \times 2 = 20\ \mu\text{l}$) and count cells using a hemocytometer (Step I.10).
- Transfer the medium containing cells into a 15 ml tubes.
- Centrifuge for 5 min at 500G.
- Freez cells

- **For plate n°2 and plate n°3**, when cells are at 70-80% confluent transfer cells into differentiation medium (2% HS-DMEM):
- Put differentiation medium in 37°C water bath for 30 min.
- Prepare Pasteur pipettes.
- Aspirate the growth medium by using a pasture pipette.
- Rinse the cells with cold PBS (2 ml).
- Add 2 ml of differentiation medium in each well.
- Incubate cells at 37°C and 5% CO₂.

[Day 3]: Change differentiation medium for plate n° 3

- Put differentiation medium in 37°C water bath for 30 min.
- Prepare Pasteur and 10 ml pipettes.
- Aspirate the differentiation medium.
- Add 2 ml of new differentiation medium in each well.
- Incubate cells at 37°C and 5% CO₂.

[Day 4]: Change differentiation medium for plate n° 2 and plate n° 3.

- For steps: see day 3.

[Day 5]: Change differentiation medium for plate n° 3.

- For steps: see day 3.

[Day 6]: Cell fixation and observation under light microscopy

- Rinse the cells twice with cold PBS buffer.
- Fix the cells for 5 min in methanol.
- Stain in hematoxylin for 5 -15 min.
- Rinse cells with DW.
- Fix the cells by adding some drops of PBS-glycerol (v / v) (Face that contains cells must be below).
- Observe the preparations under the light microscopy.

6 Effects of exogenous SPARC inhibition/addition on morphological analysis

6.1 Reagents preparation

- **10%FBS-DMEM** (Step I.4)
- **1X gelatin** (LifeTechnology cat#S-006-100)
- **2% HS-DMEM** (Step II.3)
- **1X PBS** (Step I.4)
- **Anti-SPARC antibody:** (R&D system Cat#MAB942) **(0.5 mg/ml PBS)**
- **Stock solution of recombinant SPARC protein (rSPARC):** (R&D system Cat#942-SP-050) **(100 µg/ml PBS)**
- **Methanol**
- **Hematoxylin solution according to Delafield** (Sigma Cat#03971)
- **Glycerol in PBS (v/v)** (Step II.3)

6.2 Procedure of morphological analysis:

[Day 1]: Seed cells

- **One day before adding anti-SPARC antibody or rSPARC,** seed 4×10^4 cells in 24-well plates as following :

- Take small slide
- Make it in wells
- Add 1 ml of 1X gelatin
- Incubate at 37°C for 30 min
- Aspirate gelatin
- Add growth medium and cells
- Incubate over night

► **We need in total : 28×10^4 cells**

[Day 2]: Culture cells in different conditions

- In 24-well plates, culture cells in different conditions (See table below).
- For well 1 to 6 : when cells are at 70-80% confluent transfer cells into differentiation medium (2% HS-DMEM):
- Put differentiation medium in 37°C water bath for 30 min.
- Prepare Pasteur pipettes.
- Aspirate the growth medium by using a pasture pipette.
- Rinse the cells with cold PBS (1 ml).
- Add 1 ml of differentiation medium in each well with different conditions (See table).
- Incubate cells at 37°C and 5% CO₂.

	Well 1	Well 2	Well 3	Well 4	Well 5	Well 6	Well 7
Growth medium (ml)	-	-	-	-	-	-	1
Differentiation medium (DM) (ml)	1	1	1	1	1	1	-
PBS (μl)	-	40	-	30	20	-	-
Anti-SPARC antibody (μl) at 0.5 mg/ml	-	-	20	-	-	-	-
rSPARC (μl) at 100 μg/ml	-	-	-	10	20	40	-
Final concentration of anti-SPARC antibody (μg/ml)	-	-	10	-	-	-	-
Final concentration of rSPARC (μg/ml)	-	-	-	1	2	4	-

For 1 experiment: **10 μg × 3 changes of medium = 30 μg** of anti-SPARC antibody.

For 3 repetitions: **30 μg × 3 repetitions = 90 μg** of anti-SPARC antibody.

For 1 experiment: **7 μg × 2 changes of medium = 14 μg** of rSPARC.

For 3 repetitions: **14 μg × 3 = 42 μg** of rSPARC.

[Day 4]: Change growth medium and differentiation medium

- Put growth medium (10% FBS-DMEM) and differentiation medium (2% HS-DMEM) in 37°C water bath for 30 min.
- Prepare Pasteur pipettes.
- Aspirate growth medium using Pasteur pipettes.
- Aspirate differentiation medium using an other Pasteur pipettes.
- Add 1 ml of new growth medium or differentiation medium with other reagents in each well. (As mentioned in the table above).
- Incubate cells at 37°C and 5% CO₂.
- Culture cells for 4 days (*Lim et al., 2000*) to 6 days.

[Day 6]: Change growth medium and differentiation medium

- Change differentiation medium and growth medium.

[Day 7]: Cell fixation and observation under light microscopy

- Rinse the cells twice with cold PBS buffer.
- Fix the cells for 5 min in methanol.
- Stain in hematoxylin for 5 -15 min.
- Rinse cells with DW.
- Fix the cells by adding some drops of PBS-glycerol (v / v).
- Observe the preparations under the light microscope

Fusion:

- After staining with haematoxylin, the nuclei present in myoblasts and in myotubes were counted.

$$\text{Fusion (\%)} = \frac{\text{Number of nuclei present in myotubes}}{\text{Total number of nuclei (myoblasts + myotubes)}} \times 100.$$

7 Effect of induction/ inhibition of SPARC on ECM and mitochondrial proteins

7.1 Reagents preparation

- **10% FBS-DMEM** (Step I.4)
- **2% HS-DMEM** (Step II.3)
- **1X PBS** (Step I.4)
- **0.05% trypsin-EDTA solution** (Step I.4)
- **Anti-SPARC antibody:** (R&D system Cat#MAB942) **(0.5 mg/ml PBS)**
- **Stock solution of recombinant SPARC protein (rSPARC):** (R&D system Cat#942-SP-050) **(100 µg/ml PBS)**
- **Protease inhibitors cocktail** (Sigma, Cat# R0278 store at – 20 °C)
- **RIPA buffer** (Sigma, Cat# R0278 store at 4 °C)
- **Protein Assay Kit II** (Bio-Rad, Cat# 500-0002)
- **COL1A1** (Santa Cruz Biotechnology, Cat# sc-8784-R store at 4 °C)
- **COL1A2** (Santa Cruz Biotechnology, Cat# sc-28654 store at 4 °C)
- **Fibronectin** (Santa Cruz Biotechnology, Cat# sc-9068 store at 4 °C)
- **Myogenin** (Santa Cruz Biotechnology, Cat# sc-12732 store at 4 °C)
- **Reagents for usual western**
- **MitoProfile Total OXPHOS** (Abcam, Cat# ab110413 store at 4 °C)
 - **Mouse TrueBlot Western Blot Kit** (Cedarlane, Cat# 88-8887-31 store at 4 °C)
 - **TrueBlot Assay (20X):**
 - ♠ For 100 ml of TrueBlot Assay 1X: take 5 ml of TrueBlot Assay 20X + 95 ml DW.
 - **TrueBlot Blocker (5% milk):**
 - ♠ Take 5 g of TrueBlot Blocker.
 - ♠ Add 100 ml of of TrueBlot Assay 1X.
- **CAPS transfer buffer** (See bellow)

7.2 Procedure

[Day 1]: Seed cells

- **One day before adding anti-SPARC antibody or rSPARC**, trypsinize and plate 4×10^5 of cells in 12-well plates.

[Day 2]: Culture cells in different conditions

- In 12-well plates, culture 4×10^5 of cells in different conditions (See table and picture below).
- ▶ **We need in total: 16×10^5 cells**
- For well 1 to 6 : when cells are at 70-80% confluent transfer cells into differentiation medium (2% HS-DMEM):
- Put differentiation medium in 37°C water bath for 30 min.
- Prepare Pasteur pipettes.
- Aspirate the growth medium by using a pasture pipette.
- Rinse the cells with cold PBS (2 ml).
- Add 2 ml of differentiation medium in each well with different conditions (See table).
- Incubate cells at 37°C and 5% CO₂.

	Well 1	Well 2	Well 3	Well 4
Differentiation medium (ml)	2	2	2	2
PBS (μl)	80	40	40	-
Anti-SPARC antibody (μl) at 0.5 mg/ml	-	40	-	40
rSPARC (μl) at 100 μg/ml	-	-	40	40
Final concentration of anti-SPARC antibody (μg/ml)	-	10	-	10
Final concentration of rSPARC (μg/ml)	-	-	2	20

For 1 experiment: **20 μg × 3 changes of medium = 60 μg** of anti-SPARC antibody.

For 3 repetitions: **60 μg × 3 repetitions = 180 μg** of anti-SPARC antibody.

For 1 experiment: **4 μg × 3 changes of medium = 12 μg** of rSPARC.

For 3 repetitions: **12 μg × 3 = 36 μg** of rSPARC.

[Day 4]: Change differentiation medium

- Put differentiation medium in 37°C water bath for 30 min.
- Prepare Pasteur pipettes.
- Aspirate the differentiation medium.
- Add 2 ml of new differentiation medium with other reagents in each well. (As mentioned in the table and picture above).
- Incubate cells at 37°C and 5% CO₂.
- Culture cells for 4 days. (*Ki Lim et al., 2000*) to 6 days.

[Day 6]: Change differentiation medium

- Change differentiation medium.

[Day 7]: Collect cells/ Protein extraction

Collect proteins from adherent cells :

- Put centrifuge at 4°C.
- Wash the cell 2X with 1 ml PBS/well (6-well plate) on ice.
- Add 0.5 ml of RIPA buffer sequentially to each well of the same experimental condition and scrape (detach) cells on ice by using cell lifter (VWR-Canlab Cat# 29442-200).
- Defreeze protease inhibitors cocktail (Sigma, Cat# R0278 store at – 20 °C) at RT.
- Add 1 μl of Protease inhibitors cocktail directly in the wells.
- Add again 0.5 ml of RIPA buffer. Thus, total 1 ml of RIPA buffer per condition.
- Transfer the RIPA buffer containing cells into 2 ml tubes (Step II.7.2 well n°, date).
- Sonicate 3X for 1 sec (Fisher sonic Dismembrator model 100) with 10% amplitude (Room R4739.1), wash with DW between different experimental conditions.

- Centrifuge, 14,000 rpm, 15 min, 4 °C.
- Transfer the supernatant into 2 ml tube (Exp, condition, name, date), Store sample at 80 °C.
- Aliquot 20 µl of protein extracts to quantify. ([See Western protocol Step 2.2](#))
- Make average of the concentrations of each sample.
- **Total protein in 200 µl sample (µg) = The average protein concentrations (µg/ml) * 500 (dilution) / 5 (in 200 µl).**
- **10 µg protein (µl) = 200 * 10 / Total protein in 200 µl sample (µg).**

ECM remodeling: [See Western protocol Step 3](#)

Western conditions as follow:

Name	Protein (µg)	SDS gel (%)	Transfer (V)	Transfer (h)	Blocking (h)	1 st AB (Dilution)	1 st AB (h)	2 nd AB (h)	Note
COL1A1	30	7	90	5	1	(1/200)	ON	1	-
COL1A2	5	7	90	5	1	(1/200)	ON	1	-
Fibronectin	5	7	90	5	1	(1/200)	ON	1	-
Myogenin	5	12	75	2	1	(1/200)	ON	1	2 nd is an anti-mouse monoclonal antibody
Mitochondria (OXPHOS)	30	12	75	3	ON	(1/200)	2	2	Transfer buffer is 1X CAPS

ON: Over night.

Mitochondrial function:

Changed western conditions as follow:

- Use 10% SDS-PAGE gel
- Load 30 µg of protein per well.
- Run gel at 120 V for approximately 10 min until the dye front reaches to the bottom of the stoking gel. Then, increase voltage to **150 V for approximately 60 min** until the dye front reaches to the bottom of the running gel.
- **After electrophoresis is finished the gel should be soaked in CAPS transfer buffer for 30 min before assembling the transfer sandwich.**

Using the CAPS buffer transfer system is highly recommended.

CAPS Buffer, 1 L

10 mM 3-(cyclohexylamino)-1-propanesulfonic acid, 10% methanol (pH 11.0)

CAPS	2.21 g
diH ₂ O	500 ml
Methanol	100 ml

Adjust volume to 1 L with diH₂O.

Measure the pH and adjust as needed with NaOH.

Or**► To prepare 1 liter of a 10X CAPS buffer for stock solution:**

- Place 900 ml Milli-Q water in a 1 liter beaker; insert stir bar
- Dissolve 22.13 gm CAPS (Sigma C2632, RT) in water
- Titrate dropwise with 10 M NaOH to final pH 11.0
- Transfer to a 1000 ml graduated cylinder
- Add Milli-Q water to 1000 ml
- Store at 4C

► To prepare 1 liter of working solution:

- Measure 800 ml of Milli-Q water into a 1000 ml graduated cylinder
 - Add 100 ml of 10X stock CAPS buffer
 - Add 100 ml 100% HPLC grade methanol
 - Mix well
 - Store at RT
- **Transfer at 75 V for 3 h at 4°C using CAPS transfer buffer.**
 - **Remove the membrane and soak it in CAPS transfer buffer**
 - **Under chemical hood, place the membrane in TrueBlot enhancer solution and soak for 2 min, then wash with PBST.**
 - **RedAlert staining.**
 - **Blocking: Place the membrane in 10 ml of 5% TrueBlot Blocker in TrueBlot Assay Buffer (enough to cover the membrane) and incubate overnight at 4°C on a rocking platform.**
 - **Remove the blocking buffer and wash the membrane for 10 min in PBST.**
 - **Use the primary mouse immunoblotting antibody (40 uL) in 10 mL of TrueBlot Blocker at concentration of 6 µg/ml (The antibody cocktail 1.5 mg/ml should be diluted 250x).**
 - **Incubate the blot with primary antibody for at least 2 h at RT on rocking platform.**
 - **Wash the blot at least 3 times in PBST, each wash for a minimum of 10 min each.**
 - **Prepare the secondary antibody Mouse IgG TrueBlot® at a 1:1,000 dilution in the Blocking.**
 - **Incubate the blot with the Trueblot secondary antibody for 2 h at room RT on a rocking platform.**
 - **Wash the membrane 3X in PBST solution for 10 min.**

- **The blot must be rinsed in PBS to remove any Tween-20, which can be inhibitory to the detection method.**

8. Effect of induction/ inhibition of SPARC on ECM and mitochondrial function during myoblasts proliferation:

8.1 Reagents preparation

- **10% FBS-DMEM** (Step I.4)
- **1X PBS** (Step I.4)
- **0.05% trypsin-EDTA solution** (Step I.4)
- **Anti-SPARC antibody:** (R&D system Cat#MAB942) **(0.5 mg/ml PBS)**
- **Stock solution of recombinant SPARC protein (rSPARC):** (R&D system Cat#942-SP-050) **(100 µg/ml PBS)**
- **Protease inhibitors cocktail** (Sigma, Cat# R0278 store at – 20 °C)
- **RIPA buffer** (Sigma, Cat# R0278 store at 4 °C)
- **Protein Assay Kit II** (Bio-Rad, Cat# 500-0002)
- **COL1A1** (Santa Cruz Biotechnology, Cat# sc-8784-R store at 4 °C)
- **COL1A2** (Santa Cruz Biotechnology, Cat# sc-28654 store at 4 °C)
- **Fibronectin** (Santa Cruz Biotechnology, Cat# sc-9068 store at 4 °C)
- **Myogenin** (Santa Cruz Biotechnology, Cat# sc-12732 store at 4 °C)
- **Reagents for usual western**
- **MitoProfile Total OXPHOS** (Abcam, Cat# ab110413 store at 4 °C)
- **Mouse TrueBlot Western Blot Kit**(Cedarlane, Cat# 88-8887-31 store at 4 °C)
 - **5% TrueBlot Blocker** (Step II.7.1)
 - **TrueBlot Super Block** (Step II.7.1)
- **CAPS transfer buffer** (Step II.7.1)

8.2 Procedure

[Day 1]: Seed cells

- **One day before adding anti-SPARC antibody or rSPARC**, trypsinize cells and plate 4×10^5 of cells in 12-well plates.

[Day 2]: Culture cells in different conditions

- In 12-well plates, culture 4×10^5 of cells in different conditions (See table and picture below).
- ▶ **We need in total: 16×10^5 cells**
- Put growth medium in 37°C water bath for 30 min.
- Prepare Pasteur pipettes.
- Aspirate the old growth medium by using a pasture pipette.
- Rinse the cells with cold PBS (2 ml).
- Add 2 ml of new growth medium in each well with different conditions (See table).
- Incubate cells at 37°C and 5% CO₂.

	Well 1	Well 2	Well 3	Well 4
Differentiation medium (ml)	2	2	2	2
PBS (µl)	320	160	160	-
Anti-SPARC antibody (µl) at 0.5 mg/ml	-	160	-	160
rSPARC (µl) at 100 µg/ml	-	-	160	160
Final concentration of anti-SPARC antibody (µg/ml)	-	40	-	40
Final concentration of rSPARC (µg/ml)	-	-	8	8

[Day 4]: Collect cells/ Protein extraction

Collect proteins from adherent cells :

- Put centrifuge at 4°C.
- Wash the cell 2X with 1 ml PBS/well (6-well plate) on ice.
- Add 0.5 ml of RIPA buffer sequentially to each well of the same experimental condition and scrape (detach) cells on ice by using cell lifter (VWR-Canlab Cat# 29442-200).
- Defreeze protease inhibitors cocktail (Sigma, Cat# R0278 store at – 20 °C) at RT.
- Add 1 µl of Protease inhibitors cocktail directly in the wells.
- Add again 0.5 ml of RIPA buffer. Thus, total 1 ml of RIPA buffer per condition.
- Transfer the RIPA buffer containing cells into 2 ml tubes (Step II.9.2 well n°, date).
- Sonicate 3X for 1 sec (Fisher sonic Dismembrator model 100) with 10% amplitude (Room R4739.1), wash with DW between different experimental conditions.
- Centrifuge, 14,000 rpm, 15 min, 4 °C.
- Transfer the supernatant into 2 ml tube (Exp, condition, name, date), Store sample at -80 °C.
- Aliquot 20 µl of protein extracts to quantify.

ECM remodeling:

- COL1A1 and Fn levels will be measured by western blot at **48 h** after the proliferation with or without inhibition/addition of SPARC (See Step II.7.2).

Mitochondrial function:

- The expression of mitochondria OXPHOS proteins will be measured **48 h** after the proliferation with or without inhibition/addition of SPARC (See Step II.7.2).

9. Effect of induction/ inhibition of SPARC on the expression of ECM and mitochondrial proteins in the myotubes

► Set up the number of cells, kind of plates and days for the differentiation:

9.1 Reagents preparation:

- **10% FBS-DMEM** (Step I.4)
- **2% HS-DMEM** (II.3)
- **1X PBS** (Step I.4)
- **0.05% trypsin-EDTA solution** (Step I.4)
- **Protease inhibitors cocktail** (Sigma, Cat# R0278 store at – 20 °C)
- **RIPA buffer** (Sigma, Cat# R0278 store at 4 °C)
- **Protein Assay Kit II** (Bio-Rad, Cat# 500-0002)
- **COL1A1** (Santa Cruz Biotechnology, Cat# sc-8784-R store at 4 °C)
- **Fibronectin** (Santa Cruz Biotechnology, Cat# sc-9068 store at 4 °C)
- **SDH** (Santa Cruz Biotechnology, Cat# sc-377302 or sc-390381 store at 4 °C)
- **Reagents for usual western**

9.2 Procedure :

[Day 1]: Seed cells

- **Trypsinize and plate different number of cells** (in growth medium: 10% FBS-DMEM) as shown in tables below **in 2X: 6-well, 12- well and 24-well plates.**
- **Use an other 6-well plate**

6-well plate:

Well	Number of cells
1	8×10^4
2	9×10^4
3	10^5
4	2×10^5
5	3×10^5

12-well plate:

Well	Number of cells
1	4×10^4
2	5×10^4
3	6×10^4
4	7×10^4
5	8×10^4
6	10^5

24-well plate:

Well	Number of cells
1	10^4
2	2×10^4
3	3×10^4
4	4×10^4
5	5×10^4

Other 6-well plate:

Well	Number of cells
1	3×10^4
2	3×10^4

[Day 3]: Culture cells in differentiation medium

- When cells are at 70-80% confluent transfer cells into differentiation medium (2% HS-DMEM):
- Put differentiation medium in 37°C water bath for 30 min.
- Prepare Pasteur pipettes.
- Aspirate the growth medium by using a pasture pipette.
- Rinse the cells with cold PBS (2 ml).
- Add 0.5, 1 and 2 ml of differentiation medium in 24-well, 12-well and 6-well plates, successively.
- Incubate cells at 37°C and 5% CO₂.

NB: Keep cells from the 3rd 6-well plate (contains 2 wells only) in GM and change GM for the first well only.

[Day 4]: Change differentiation medium for the first 24-well, 12-well and 6-well plate

Change growth medium to differentiation medium for 6-well plate (with 2 wells only)

- Put differentiation medium in 37°C water bath for 30 min.
- Prepare Pasteur pipettes.
- Aspirate the differentiation medium.
- Add 3 ml of new differentiation medium with other reagents in each well.
- Incubate cells at 37°C and 5% CO₂.

[Day 5]: Change differentiation medium for all the plates and for the second well of 6-well plate only**[Day 6]: Change differentiation medium for the first 24-well, 12-well and 6-well plate and for the 6-well plate with 2 wells.****[Day 7]: Change differentiation medium for all the plates and for 2nd well of 6-well plate**

[Day 8]: Change differentiation medium for the first 24-well, 12-well and 6-well plate and for the 6-well plate with 2 wells.

[Day 9]: Change differentiation medium for all the plates and for 2nd well of 6-well plate

[Day 10]: Change differentiation medium for the first 24-well, 12-well and 6-well plate and for the 6-well plate with 2 wells.

[Day 10]: Protein extraction.

Remarks:

6-well plate:

Well	Number of cells	% Confluency
1	8×10^4	60
2	9×10^4	70
3	10^5	80
4	2×10^5	90
5	3×10^5	100

12-well plate:

Well	Number of cells	% Confluency
1	4×10^4	60
2	5×10^4	70
3	6×10^4	80
4	7×10^4	90
5	8×10^4	90
6	10^5	100

24-well plate:

Well	Number of cells	% Confluency
1	10^4	40
2	2×10^4	60
3	3×10^4	70
4	4×10^4	80
5	5×10^4	90

Other 6-well plate:

Well	Number of cells	% Confluency
1	3×10^4	80
2	3×10^4	80

- Partial detachment starts on the **6th day** of the differentiation.
- Detachment is **less** in wells that DM is changed **every day**.
- It's better to **add SPARC** on the **5th day** of the differentiation.
- The adequate number of cells for each plate is mentioned **in red** (see tables above).

10. Effect of induction/ inhibition of SPARC on the expression of ECM and mitochondrial protein in the myotubes

► Set up the adequate concentration of rSPARC and anti-SPARC antibody:

10.1 Reagents preparation:

- **10% FBS-DMEM** (Step I.4)
- **2% HS-DMEM** (II.3)
- **1X PBS** (Step I.4)
- **0.05% trypsin-EDTA solution** (Step I.4)
- **Anti-SPARC antibody:** (R&D system Cat#MAB942) **(0.5 mg/ml PBS)**
- **Stock solution of recombinant SPARC protein (rSPARC):** (R&D system Cat#942-SP-050) **(100 µg/ml PBS)**
- **Protease inhibitors cocktail** (Sigma, Cat# R0278 store at – 20 °C)
- **RIPA buffer** (Sigma, Cat# R0278 store at 4 °C)
- **Protein Assay Kit II** (Bio-Rad, Cat# 500-0002)
- **COL1A1** (Santa Cruz Biotechnology, Cat# sc-8784-R store at 4 °C)
- **Fibronectin** (Santa Cruz Biotechnology, Cat# sc-9068 store at 4 °C)
- **SDH** (Santa Cruz Biotechnology, Cat# sc-377302 or sc-390381 store at 4 °C)
- **COX 1** (Santa Cruz Biotechnology, Cat# sc-48143)
- **AMPK** (Santa Cruz Biotechnology, Cat#25792 sc-at 4 °C)
- **Thr172** (Santa Cruz Biotechnology, Cat# sc-33524 store at 4 °C)
- **Reagents for usual western**

10.1 Procedure :

[Day 1]: Seed cells

- **Trypsinize cells and plate 2×10^4 cells/well** (in growth medium: 10% FBS-DMEM) **in 3X 24-well plates.**

[Day 3]: Culture cells in differentiation medium

- **When cells are at 70-80% confluent transfer cells into differentiation medium (2% HS-DMEM):**
- **Put differentiation medium in 37°C water bath for 30 min.**
- **Prepare Pasteur pipettes.**
- **Aspirate the growth medium by using a pasture pipette.**
- **Rinse the cells with cold PBS (0.5 ml)**
- **Add 0.5 ml of differentiation medium in each well.**
- **Incubate cells at 37°C and 5% CO₂.**

[Day 4]: Change differentiation medium

- **Put differentiation medium in 37°C water bath for 30 min.**
- **Prepare Pasteur pipettes.**
- **Aspirate the differentiation medium.**
- **Add 0.5 ml of new differentiation medium with other reagents in each well.**

- Incubate cells at 37°C and 5% CO₂.

[Day 5]: Change differentiation medium

[Day 6]: Change differentiation medium

[Day 7]: Change differentiation medium

[Day 8]: Culture cells in different conditions

- Change differentiation medium.
- Culture cells in different conditions. (see table below)

	Well 1	Well 2	Well 3	Well 4	Well 5	Well 6
Differentiation medium (ml)	0.5	0.5	0.5	0.5	0.5	0.5
PBS (µl)	80	70 (10)	60 (20)	60 (20)	40	80
Anti-SPARC antibody (µl) at 0.5g/ml	-	10	20	-	-	-
rSPARC (µl) at 100 µg/ml (µg/ml)	-	-	-	20	40	80
Final concentration of anti- SPARC antibody	-	10	20	-	-	-
Final concentration of rSPARC (µg/ml)	-	-	-	4	8	16

NB: The volume of PBS that I took for my experiment is in red, the real volume must be the one in black.

[Day 9]: Collect cells/ Protein extraction for plate n° 1

Collect proteins from adherent cells :

- Put centrifuge at 4°C.
 - Wash the cell 2X with 0.5 ml PBS/well (24-well plate) on ice.
 - Add 0.25 ml of RIPA buffer sequentially to each well of the same experimental condition and scrape (detach) cells on ice by using cell lifter (VWR-Canlab Cat# 29442-200).
 - Defreeze protease inhibitors cocktail (Sigma, Cat# R0278 store at – 20 °C) at RT.
 - Add 1 µl of protease inhibitors cocktail directly in the wells.
 - Add again 0.25 ml of RIPA buffer. Thus, total 0.5 ml of RIPA buffer per condition.
 - Transfer the RIPA buffer containing cells into 2 ml tubes (Step II.11.2 well n°, date).
 - Sonicate 3X for 1 sec (Fisher sonic Dismembrator model 100) with 10% amplitude (Room R4739.1), wash with DW between different experimental conditions.
 - Centrifuge, 14,000 rpm, 15 min, 4 °C.
 - Transfer the supernatant into 2 ml tube (Exp, condition, name, date), Store sample at -80 °C.
 - Dilute cell culture lysate 2X: 10 µl of sample + 10 µl of DW in 1.7 ml tubes.
 - *Protein quantification should make with duplicate. Do not forget to make a pooled sample (western control sample) for protein quantification.
 - Aliquot 20 µl of protein extracts to quantify with duplicate.
- Make average of the concentrations of each sample.

- **Total protein in 200 µl sample (µg) = The average protein concentrations (µg/ml) * 500 (dilution) / 5 (in 200 µl).**
- **10 µg protein (µl) = 200 * 10 / Total protein in 200 µl sample (µg).**

[Day 10]: Collect cells/ Protein extraction for plate n° 2 and change medium for plate n° 3

- **Change differentiation medium.**
- **Culture cells in different conditions.** (see table below)

	Well 1	Well 2	Well 3	Well 4	Well 5	Well 6
Differentiation medium (ml)	0.5	0.5	0.5	0.5	0.5	0.5
PBS (µl)	80	70 (10)	60 (20)	60 (20)	40	80
Anti-SPARC antibody (µl) at 0.5g/ml	-	10	20	-	-	-
rSPARC (µl) at 100 µg/ml (µg/ml)	-	-	-	20	40	80
Final concentration of anti-SPARC antibody (µg/ml)	-	10	20	-	-	-
Final concentration of rSPARC (µg/ml)	-	-	-	4	8	16

[Day 11]: Collect cells/ Protein extraction for plate n° 3

ECM remodeling & Mitochondrial function& AMPK expression: [See Western protocol Step 3](#)

Western conditions as follow: To complete after doing WB.

Name	Protein (ug)	SDS gel (%)	Transfer (V)	Transfer (h)	Blocking (h)	1 st AB (Dillution)	1 st AB (h)	Note
COL1A1	30	7	90	5	1	(1/200)	ON	
Fibronectin	5	7	90	5	1	(1/200)	ON	
SDHB	10	12	75	2	1	(1/200)	ON	
COX 1	30	10	90	3	1	(1/200)	ON	The 2 nd antidbody is anti-goat.
AMPK	20	10			1	(1/200)	ON	
Thr172	20	10			1	(1/200)	ON	

NB: Final concentration of rSPARC is 4 µg/ml, anti-SPARC antibody, we need to **set up** its concentration.

11. Effect of induction/ inhibition of SPARC on the expression of ECM and mitochondrial protein in the myotubes

► Set up the adequate concentration of anti-SPARC antibody:

11.1 Reagents preparation:

- **10% FBS-DMEM** (Step I.4)
- **2% HS-DMEM** (II.3)
- **1X PBS** (Step I.4)
- **0.05% trypsin-EDTA solution** (Step I.4)
- **Anti-SPARC antibody:** (R&D system Cat#MAB942) **(0.5 mg/ml PBS)**
- **Stock solution of recombinant SPARC protein (rSPARC):** (R&D system Cat#942-SP-050) **(100 µg/ml PBS)**
- **Protease inhibitors cocktail** (Sigma, Cat# R0278 store at – 20 °C)
- **RIPA buffer** (Sigma, Cat# R0278 store at 4 °C)
- **Protein Assay Kit II** (Bio-Rad, Cat# 500-0002)
- **COL1A1** (Santa Cruz Biotechnology, Cat# sc-8784-R store at 4 °C)
- **Fibronectin** (Santa Cruz Biotechnology, Cat# sc-9068 store at 4 °C)
- **SDH** (Santa Cruz Biotechnology, Cat# sc-377302 or sc-390381 store at 4 °C)
- **COX 1** (Santa Cruz Biotechnology, Cat# sc-48143)
- **AMPK** (Santa Cruz Biotechnology, Cat#25792 sc-at 4 °C)
- **Thr172** (Santa Cruz Biotechnology, Cat# sc-33524 store at 4 °C)
- **Reagents for usual western**

11.2 Procedure :

[Day 1]: Seed cells

- Trypsinize cells and plate 2×10^4 cells/well (in growth medium: 10% FBS-DMEM) in 2X 24-well plates. **We need in total: $2 \times 10^4 \times 4$ conditions $\times$ 2 plates = 16×10^4**

[Day 3]: Culture cells in differentiation medium

- When cells are at 70-80% confluent transfer cells into differentiation medium (2% HS-DMEM):
- Put differentiation medium in 37°C water bath for 30 min.
- Prepare Pasteur pipettes.
- Aspirate the growth medium by using a pasture pipette.
- Rinse the cells with cold PBS (0.5 ml)
- Add 0.5 ml of differentiation medium in each well.
- Incubate cells at 37°C and 5% CO₂.

[Day 4]: Change differentiation medium

- Put differentiation medium in 37°C water bath for 30 min.
- Prepare Pasteur pipettes.
- Aspirate the differentiation medium.
- Add 0.5 ml of new differentiation medium with other reagents in each well.

- Incubate cells at 37°C and 5% CO₂.

[Day 5]: Change differentiation medium

[Day 6]: Change differentiation medium

[Day 7]: Change differentiation medium

[Day 8]: Culture cells in different conditions

- Change differentiation medium.
- Culture cells in different conditions. (see table below)

	Well 1	Well 2	Well 3	Well 4
Differentiation medium (ml)	0.5	0.5	0.5	0.5
PBS (µl)	80	70	40	60
Anti-SPARC antibody (µl) at 0.5g/ml	-	10	40	-
rSPARC (µl) at 100 µg/ml (µg/ml)	-	-	-	20
Final concentration of anti-SPARC antibody (µg/ml)	-	10	40	-
Final concentration of rSPARC (µg/ml)	-	-	-	4

[Day 10]: Collect cells/ Protein extraction for plate n° 1 and change medium for plate n° 2

Collect proteins from adherent cells :

- Put centrifuge at 4°C.
 - Wash the cell 2X with 0.5 ml PBS/well (24-well plate) on ice.
 - Add 0.25 ml of RIPA buffer sequentially to each well of the same experimental condition and scrape (detach) cells on ice by using cell lifter (VWR-Canlab Cat# 29442-200).
 - Defreeze protease inhibitors cocktail (Sigma, Cat# R0278 store at – 20 °C) at RT.
 - Add 1 µl of protease inhibitors cocktail directly in the wells.
 - Add again 0.25 ml of RIPA buffer. Thus, total 0.5 ml of RIPA buffer per condition.
 - Transfer the RIPA buffer containing cells into 2 ml tubes (Step II.12.2 well n°, date).
 - Sonicate 3X for 1 sec (Fisher sonic Dismembrator model 100) with 10% amplitude (Room R4739.1), wash with DW between different experimental conditions.
 - Centrifuge, 14,000 rpm, 15 min, 4 °C.
 - Transfer the supernatant into 2 ml tube (Exp, condition, name, date), Store sample at -80 °C.
 - Dilute cell culture lysate 2X: 10 µl of sample + 10 µl of DW in 1.7 ml tubes.
 - ***Protein quantification should make with duplicate. Do not forget to make a pooled sample (western control sample) for protein quantification.**
 - Aliquot 20 µl of protein extracts to quantify with duplicate.
- Make average of the concentrations of each sample.
 - **Total protein in 200 µl sample (µg) = The average protein concentrations (µg/ml) * 500 (dilution) / 5 (in 200 µl).**

- **10 µg protein (µl) = 200 * 10 / Total protein in 200 µl sample (µg).**

[Day 12]: Collect cells/ Protein extraction for plate n° 2

ECM remodeling & Mitochondrial function& AMPK expression: See Western protocol Step 3

Western conditions as follow: To complete after doing WB.

Name	Protein (ug)	SDS gel (%)	Transfer (V)	Transfer (h)	Blocking (h)	1 st AB (Dillution)	1 st AB (h)	Note
COL1A1	30	7	90	5	1	(1/200)	ON	
Fibronectin	5	7	90	5	1	(1/200)	ON	
SDHB	10	12	75	2	1	(1/200)	ON	
COX 1	30	10	90	3	1	(1/200)	ON	The 2 nd antidbody is anti-goat.
AMPK	20	10			1	(1/200)	ON	
Thr172	20	10			1	(1/200)	ON	

Appendix 1: Venor®GeM Mycoplasma Detection Kit

1. For conventional PCR

Cat.-No. 11-1025 25 Tests

Cat.-No. 11-1050 50 Tests

Cat.-No. 11-1100 100 Tests

Cat.-No. 11-1250 250 Tests

Description

Venor®GeM employs PCR technology for rapid and reliable detection of all relevant mycoplasma species in various in situ biological including cell cultures and virus-stocks. For fast, reliable and time-saving routine monitoring!

Sensitive

Detection of 1.5 copies/μl of mycoplasma DNA that corresponds to 3 mycoplasma per sample volume. $LOD_{95\%} = 4.5$ copies/μl.

Specific

Venor®GeM is specific for more than 79 mycoplasma species.

Reliable

The kit contains internal control DNA and positive control DNA to optimize reproducibility and minimize operator-dependent variability of results.

Validated

Venor®GeM was intensively validated according to the guidelines of the European Pharmacopoeia, section 2.6.7.

Simple

Templates for PCR analysis are prepared by boiling the supernatant of cell cultures or other sample material.

Clear

A clear and easily interpretable result is obtained with one PCR reaction.

Quality Control Considerations

Mycoplasma are often found as common contaminants in cell cultures, virus stocks, and other cell-derived biological. Mycoplasma contamination is a recurring problem that can cause non-reproducible results in research and productivity losses of up to 50% in industry, which can be avoided by routine diagnostic testing. Nowadays, testing for mycoplasma infection is an essential requirement in reaching quality control standards for all cell culture laboratories. It is necessary to demonstrate quality in laboratories concerned with the production and maintenance of cell lines used in manufacturing various biological and in research laboratories to validate their results.

Positive and Internal Controls

Venor® GeM incorporates the highly-sensitive PCR technology as its method for detecting contaminating mycoplasma, which also offers outstanding characteristics over other techniques and products used for mycoplasma safety. The kit includes positive and internal controls. The internal control confirms proper amplification by acting as an alternative template for the PCR primers. Non-infectious *Mycoplasma orale* DNA is used as a positive control.

Kit Components

- Primer sets and nucleotides at optimized concentrations
- 10x reaction buffer optimized for many *Taq* Polymerases
- Positive control for easy result verification
- Internal control to ensure accurate testing
- Instruction manual & storage box

We highly recommend our reliable hot-start MB Taq DNA Polymerase for PCR detection of mycoplasma (Cat.- No. 53- 0200).

Venor®GeM kit components are lyophilized and at +2 to +8 °C maintainable for at least 12 months. Primer and nucleotides are aliquoted (à 25 tests) to guarantee consistent quality. After rehydratisation the reagents must be stored at -18 °C.

2. VenoräGeM Mycoplasma Detection Kit, PCR-based

Catalog Number **MP0025**

Storage Temperature 2–8 °C

Product Description

- The VenoräGeM Mycoplasma Detection Kit utilizes the polymerase chain reaction (PCR), which was established as the method of choice for highest sensitivity in the detection of *Mycoplasma* and *Acholeplasma* contamination in cell cultures and other cell culture derived biologicals. Detection requires as little as 1–5 fg of mycoplasma DNA corresponding to 2–5 mycoplasma per sample volume.
- The primer set is specific to the highly conserved Rna operon, or more specifically, the 16S Rna coding region in the mycoplasma genome.
- This allows for detection of all *Mycoplasma*, *Acholeplasma*, and *Ureaplasma* species tested so far, which are usually encountered as contaminants in cell cultures.
- Eukaryotic and bacterial DNA are not amplified by this kit.
- Only one protocol is needed for the detection of all mycoplasma species.
- The detection procedure can be performed within 3 hours.
- This kit also provides internal control DNA, which can be added to the reaction.
- When running the PCR with the internal control DNA, a successfully performed reaction is indicated by a distinct 191 bp band on the agarose gel.

Components

1. Red cap : P9242

- Primer set and nucleotides
- Lyophilized primers and deoxynucleotide triphosphates Datp, Dctp, Dgtp and Dttp; aliquoted for 25 reactions

2. Blue cap : P9367

- PCR 10' reaction buffer sterile, 500 ml
- 100 Mm Tris-HCl (Ph 8.5)
- 750 Mm KCl
- 30 Mm MgCl₂

3. Green cap : P9117

- Positive Control DNA non-infectious DNA-fragments of *Mycoplasma orale* genome prepared by PCR
- Lyophilized

4. Yellow cap : I0532

- Internal Control DNA
- Lyophilized plasmid DNA including mycoplasma-specific primer sequences and an internal sequence of the HTLV-I *tax* gene with a size of ~191 bp, non-infectious

Precautions and Disclaimer

This product is for R&D use only, not for drug, household, or other uses. It is not for clinical Diagnostics or testing of human samples.

Storage/Stability

- Kit components are stable during shipping at RT.
- Upon receipt, store at 2–8 °C.
- After rehydration of the primer/nucleotide mix, the positive control, and the internal control, store below –20 °C and avoid repeated freezing and thawing.
- For repeated testing of low sample numbers, primer/nucleotide mix and controls should be aliquoted after rehydration.
- By following these recommendations, the kit is stable until the expiration date stated on the label.

Procedure**Preparation of sample material**

- Cell lines should be pre-cultured in the absence of antibiotics for several days to maximize test sensitivity.
- Samples should be derived from cultures that are at 90–100% confluence.
- PCR inhibiting substances may accumulate in the medium of older cultures.
- For these sample materials a DNA extraction is strictly recommended prior to testing.
- To avoid false positive results, it is recommended to use deionized, DNA-free water, aerosol-preventive filter tips, and gloves.
- Templates for PCR analysis are prepared by boiling the supernatant of cell cultures or other biologicals for 5 min as follows :

1. Transfer 100 µl of supernatant from the test culture to a sterile microcentrifuge tube. The lid should be tightly sealed to prevent opening during heating.
2. Heat the sample supernatant at 95 °C for 5 min.
3. Briefly centrifuge (5 s) the sample, supernatant to pellet cellular debris before adding to the PCR mixture.

Rehydration of the Reagents

1. Before rehydrating the tubes, centrifuge the tubes to ensure that the lyophilized components are spun down (5 s at maximum speed).

2. Add the appropriate amount of deionized, DNA-free water:

Primer/nucleotide mix	65 µl
(per portion of 25 reactions)	

Positive control	300 µl
------------------	--------

Internal control	300 µl
------------------	--------

3. Incubate for 5 min at RT.
4. Vortex and centrifuge again.
5. Keep reagents on ice and store below –20 °C after rehydration.

Thermal Profile

1. The programming process of your cycler is explained in the instrument's manual.
2. The incubation time depends on the polymerase used. Hot start enzymes need to be activated at 94 °C. Please see polymerase data sheet for duration.

Thermal Cycle Program

1 cycle	94 °C	for 2 min
39 cycles	94 °C	for 30 s
	55 °C	for 1 min
	72 °C	for 30 s
Cool down to 4–8 °C		

The PCR Mastermix

Total volume per reaction is 25 µl. When setting up reactions, calculations should also include positive and negative controls.

1. Preparation of a PCR Mastermix

Use *Taq* polymerase (with a final MgCl₂ concentration of 3.0 mM, is detailed in Table 1. The mastermix can also be prepared for 25 reactions, aliquoted as needed and stored below –20 °C for up to 3 months.

Aliquot 23 µl of PCR mastermix into each PCR reaction tube. For other polymerase concentration

2. Add 2 µl of deionized, DNA-free water as a negative control into reactions tubes and seal to avoid contamination.
3. Add 2 µl of sample (as previously described) to PCR reaction tube per sample being tested and seal.
4. Pipette 2 µl of positive control DNA into positive control reaction tube.
5. Proceed to thermal cycling

Agarose Gel Run

1. Use 1.5 % standard agarose gel with 5 mm comb.
2. Load 5 µl of each PCR reaction, mixed with bromophenol blue loading buffer per lane.
3. Stop electrophoresis after 2 cm run distance depending on the electrophoresis chamber used e.g., run for 20 minutes at 100 V).

Results

Gel Evaluation

1. If internal control DNA was used, a distinct 191 bp band should appear in every lane indicating a successfully performed PCR. This band may fade out with increased amount of amplicons formed, caused by mycoplasma DNA loads of $>5 \times 10^6$ copies/ml.

2. No amplification of control DNA may be due to the following reasons:

- (1) Activity of *Taq* polymerase is insufficient;
- (2) Reaction buffer is not suitable for polymerase;
- (3) Control DNA tubes have not been spun down before rehydration;
- (4) Programming mistake;
- (5) Pipetting mistake.

- Before rerun of a negative and a positive control please check thermocycler protocol and pipetting scheme.

Polymerase concentration can be raised up to 2.5 U/reaction. The VenorGeM buffer can be replaced by the specific buffer provided with the polymerase; however, the magnesium concentration must then be adjusted to 3.0 mM.

- The initial concentration of positive control DNA exceeds 5×10^6 copies/ml in order to account for DNA loss resulting from repeated freeze/thaw cycles.

- With VenorGeM designed for high sensitivity and therefore, prone to nonspecific annealing, bands of various lengths that are less intensive can be produced, but do not indicate positive results. Possible primer self-annealing produces another band of 80–90 bp in length, but also does not affect the precision or results of the test.

- If the PCR of a sample is inhibited, PCR inhibitors can easily be removed from the sample by performing a DNA extraction with a commercially available kit.

Table 1.
Pipetting schemes for Preparation of PCR Mastermix

	1 reaction	3 reactions	25 reactions*
Water	15.1 µl	45.3 µl	377.5 µl
10× reaction buffer (blue cap)	2.5 µl	7.5 µl	62.5 µl
Primer/ nucleotide mix (red cap)	2.5 µl	7.5 µl	62.5 µl
Internal control (yellow cap)	2.5 µl	7.5 µl	62.5 µl
Taq Polymerase (2.5 U/µl) Catalog Number D9307	0.4 µl	1.2 µl	10 µl

* 25 reactions are the equivalent of the content of one red-capped vial.

Note: Use of other polymerases is possible; however, the PCR buffer supplied with the enzyme should be used. The final MgCl₂ concentration must be adjusted to 3.0 mM. For PCR buffers with separate MgCl₂ solutions, Table 2 may be used as a guide for the volume of the MgCl₂ solution used in a 25 µl PCR reaction.

Table 2.
Volume of Separate MgCl₂ Solutions to Prepare Other PCR Buffers

MgCl ₂ Solution Concentration	Volume of MgCl ₂ Solution per PCR reaction
25 mM	3.0 µl
50 mM	1.5 µl
100 mM	0.75 µl

Table 2.
Relevant amplicon sizes

Internal control	191 bp
<i>Mycoplasma sp.</i> (see Appendix)	~270 bp

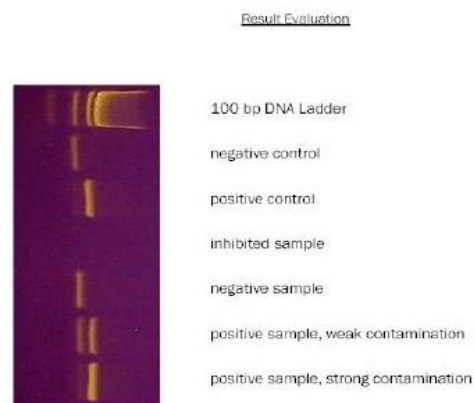
Table 3.
Results of successfully performed PCR

PCR sample	Band pattern
Negative control	Band at 191 bp
Positive control	Band at 267 bp, possibly an additional band at 191 bp

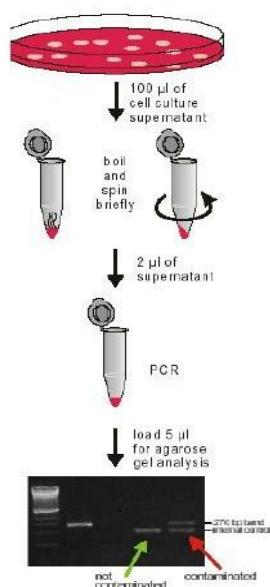
Table 4.
Interpretation of possible band patterns

Band pattern	Interpretation
Band at 191 bp	Negative sample
Bands at 270 bp and 191 bp	Mycoplasma-positive sample with weak contamination
Strong band at 270 bp	Mycoplasma-positive sample with strong contamination
No band	PCR inhibition, insufficient polymerase activity

Figure 1.



Schema of the protocol:



Appendix

Detection Range and Sizes of Amplicons

No.	species	amplicon size (bp)
1	<i>Mycoplasma pneumoniae</i> ²	277
2	<i>Mycoplasma penetrans</i>	274
3	<i>Mycoplasma pirum</i>	274
4	<i>Acholeplasma laidlawii</i> ²	273
5	<i>Mycoplasma fermentans</i>	272
6	<i>Ureaplasma urealyticum</i>	271
7	<i>Mycoplasma hyorhinis</i> ²	269
8	<i>Mycoplasma pulmonis</i>	268
9	<i>Mycoplasma falconis</i>	268
10	<i>Mycoplasma orale</i> ^{1,2}	267
11	<i>Mycoplasma arthritidis</i>	267
12	<i>Mycoplasma arginini</i>	267
13	<i>Mycoplasma spermatophilum</i>	267
14	<i>Mycoplasma opalescens</i>	267
15	<i>Mycoplasma primum</i>	267
16	<i>Mycoplasma maculosum</i>	267
17	<i>Mycoplasma bovis</i>	267
18	<i>Mycoplasma cloacale</i>	266
19	<i>Mycoplasma hyosynoviae</i>	266
20	<i>Mycoplasma synoviae</i> ²	266
21	<i>Mycoplasma salivarium</i>	266
22	<i>Mycoplasma faucium</i>	265
23	<i>Mycoplasma hominis</i>	267
24	<i>Mycoplasma genitalium</i>	267

¹ provided as positive control DNA

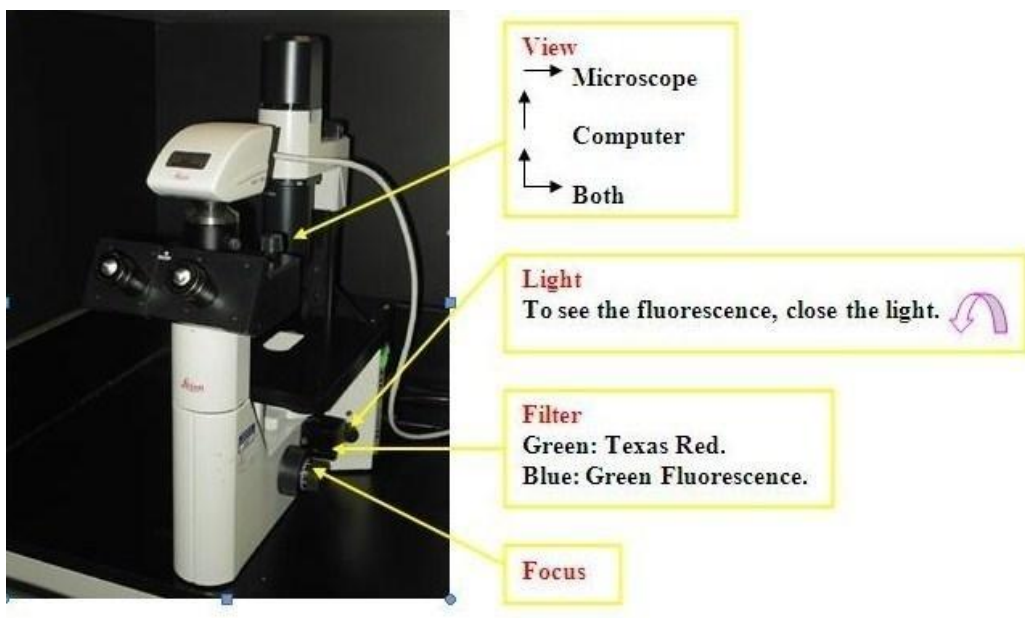
² Test strain according to European Pharmacopoeia, Suppl. 2000, 2.6.7. Mycoplasmas

Appendix 3: Fluorescence microscope

Responsible: Ms. Andrea Fournier (46315).....**to confirm**

Combination of key: press 2 and 4, then 3.....**to confirm**

1. Switch on lamp 30 min before use. The switch is on the small box beside computer.
2. Write the time and name on the sheet.
3. Switch on computer and monitor.
4. Enter User name: andrea and Pass word: andrea, and choose “CXT342A (this computer).
5. Click on “IM50” program.
6. Click on  to view the image.
- 7.
8. Click  to take a picture.
9. Save data to “DATA (E :)  **Mayomi....personal file**
10. When finished, close the lamp and write time finished.
11. Shutdown the computer and turn off the monitor.



Reference: Dr, Mayumi Yoshioka.

Appendix 4: List of instruments/ reagents**List of instruments :**

Item	Company	Cat#	Dimension	CAN\$
Fridge	Danby			
Freezer –80 °C MDFU50V	Sanyo (ESBE)		18.5picu	7700.00
Ratelier pour 24 boîtes de 2”	ESBE	CY0462		175.00
Ratelier pour 20 boîtes de 2”	ESBE	CY0452		185.00
Balance avec vitre AB204-SRS (Switzerland)	Fisher	/MEAB204S		3190.00
Eppendorf Micro Centrifuge Model 5415D with 18 X 1.5-2.0ml rotor, 115V/60Hz	VWR-Canlab	20901-051		2850.00
Boekel digital dry bath incubator for 2-block (Isotemp 125D)	Fisher	11-715-125D		750.00
Interchangeable heating blocks 0.5 ml	Fisher	11-718-23		50.00
Interchangeable heating blocks 1.5 ml	Fisher	11-718-9		50.00
Emerson Microwave 1.6ft3	4 Bourgeois		1100 watt 26 cm H	189.99
Mini vortex MV1	VWR-Canlab	33994-306		330.66
Corning scholar 171 stirrer model PC171	Fisher	11-497-22		210.00
TFE starburst stirring heads kit	Fisher	14-511-96		24.51
Chariot 3 tabl PL/Alum 36”haut	Fisher	1192671		237.25
Nalgene LDPE carboys with spigots 20 l	Fisher	02-963-6B		164.87
Gilson Pipetman® P2	MANDEL	GF-F144801		449.00
Pipetman P20	MANDEL	GF23600		389.00
Pipetman P200	MANDEL	GF23601		389.00
Pipetman P1000	MANDEL	GF23602		389.00
Pipetman Classic Large Volume Kit (P5000, P10ml)	MANDEL	GF-F167400		695.00
Pipet Helper, Brinkmann	VWR-Canlab	53573-000		79.20
Brinkmann Eppendorf Repeater Plus Pipetter	Fisher	21-380-9		425.00
Thermometer 20 Oc				13.65
Incubator Safety thermometers	VWR-Canlab	CA61222-524		64.25
Triple-Purpose Timer	VWR-Canlab	CA62344-904		29.67
Replacement Battery, AAA Alkaline	VWR-Canlab	13709-804		2.18
Isotemp water-jacketed incubator (CO ₂ incubator)	Fisher	546		
CO ₂ bombe				
CO ₂ measurement	Bacharach Instrument Company	Fire		
Biological Safety Cabinets	Nuaire Inc	NU-425-600		
Clamp				
Eppendorf multipurpose centrifuge refrigerated 5810R	VWR-Canlab	53513-812		9830.00
Eppendorf multipurpose centrifuge refrigerated 5810R Standard fixed-angle rotor, for 30 microcentrifuge tubes	VWR-Canlab	53513-816		985.00
Water bath SW22 JULABO	Fisher	9550322		3400.00
Gable Cover for Waterbath	Fisher	39968600		295.00
Pipette Sterilizing Box (23cm)	Fisher	03-475-5		40.30

Hand Tally Counter	Fisher	07-905-6		31.66
Counting chamber (Bright-Line)	Fisher	02-671-5		133.48
Phase Hemocytometer Counting chamber (Bright-Line)	Fisher	02-671-6		142.54
Microscope	Olympus	CK30		
Potable LN ₂ Storage Containers from Thermo Election	Fisher	TF8037	CMC Series 33.4L	1251.00
Smartspec 3000	Bio-Rad	170-2501		6192.00
Submicrovolume cuvette, quartz	Bio-Rad	170-2505		566.00
Fluorescence Microscope	Leica	DM1L		
Sonic Dismembrator Model 500	Fisher Scientific			
Power supply 220V	BIO-RAD	1655050		540.95
Mini-PROTEIN3 Electrophoresis Cell	BioRad	165-3322	1 mm	878.00
Mini-Trans-Blot Cell	BioRad	170-3930		662.00
Roller	BioRad	165-1279		30.00
AlphaImagerTM1220 Documentation and analysis system digital imaging system	Canberra Packard Canada			13,915.00
Sony UPD-890 Digital thermal Printer	Canberra Packard Canada			3,035.00
Kodak BioMax® Cassette, w/o screen	PerkinElmer	8209140	8x10 in	360.00
Q_RT-PCR				

List of expendable supplies

Item	Company	Cat#	Dimension	CAN\$
Pyrex brand Beaker 50 ml	Fisher	02-540G	12 pack	34.68
Pyrex brand Beaker 100 ml	Fisher	02-540H	12 pack	32.67
Pyrex brand Beaker 250 ml	Fisher	02-540K	12 pack	34.68
Pyrex brand Beaker 600 ml	Fisher	02-540M	6 pack	23.13
Pyrex brand Beaker 1000 ml	Fisher	02-540P	6 pack	44.01
Bayer 1000 ml	Tissue tek	P4110315 1308423 999009		
Cylindre gradue 50 ml	VWR-Canlab	CA24712-094	4 pack	62.98
Nalgene PP cylinders 100 ml	Fisher	08-572D		42.09
Nalgene PP cylinders 500 ml	Fisher	08-572F		74.45
Nalgene PP economy cylinders 1000 ml	Fisher	08-572-6G		94.37
Nalgene polypropylene cylinders 2000 ml	Fisher	08-572H		156.79

Pyrex brand flasks with reinforced tooled top 250 ml	Fisher	10-040F	12 pack	71.15
Pyrex brand flasks with reinforced tooled top 500 ml	Fisher	10-040H	6 pack	44.40
Pyrex brand flasks with reinforced tooled top 2000 ml	Fisher	10-040M	1	28.32
Two-Hole Rubber stoppers (size No 10)	Fisher	14-140M	5 pack	133.14
Pyrex Brand Standard Wall Tubing	Fisher	11-365-FF	25 lb	363.10
Pyrex brand Reusable laboratory bottles 250 ml	Fisher	06-414-1B	10 pack	72.61
Pyrex brand Reusable laboratory bottles 500 ml	Fisher	06-414-1C	10 pack	88.67
Pyrex brand Reusable laboratory bottles 1000 ml	Fisher	06-414-1D	10 pack	123.69
Wide mouth amber glass jar and bottles 250 ml	Fisher	05-719-294	12 pack	22.51
Polypropylene jars with screw cap 500 ml	VWR-Canlab	CA36318-858		46.45
Polypropylene straight-sided Jars, Nalgene 1000 ml 6pack	VWR-Canlab	CA36319-707		61.50
Spray bottle	Dollar shop			1.00
Fisherbrand 500 ml Easy-Squeeze Wash Bottle	Fisher	02-897-10	case of 6	60.33
Filling funnels Fisher brand power	Fisher	10-348-2D	147 mm	10.26
Microcentrifuge tube storage racks	Fisher	05-541-1	5 pack Assorted	30.87
Fisher brand interlocking four-way tube racks	Fisher	03-448-17	Assorted colors	36.40
Floating Microtube Racks, NALGENE, White	VWR-Canlab	60986-098	4 pack	132.99
Storage box for 100 microtubes	Sarstedt	95.064.997	12 units	38.40
Storage box for 81 microtubes	Sarstedt	95.064.981	12 units	38.40
NonPowder latex Medical Examination gloves size L	CHUL store	02-0011501	100	7.75
NonPowder latex Medical Examination gloves size M	CHUL store	02-0011502	100	7.75
NonPowder latex Medical Examination gloves size S	CHUL store	02-0011503	100	7.75
Powder latex Medical Examination gloves size L	CHUL store	02-0011014	100	4.84
Powder latex Medical Examination gloves size M	CHUL store	02-0011013	100	4.84
Powder latex Medical Examination gloves size S	CHUL store	02-0011012	100	4.49
Kimwipe EX-L	CHUL store	04-2021507	280	2.63
Kimtowels	CHUL store	04-2021504	200	
Parafilm M laboratory wrapping film	Fisher	13-374-12	4*250	34.67
Aluminium paper				

VWRbrand weight paper	VWR-Canlab	CA12578-121	3*3	25.16
Weighing Dishes, 1 5/8*1 5/8*5/16	VWR-Canlab	12577-005	500	43.16
Weighing Dishes, 3 1/2*3 1/2*1	VWR-Canlab	12577-027	500	51.45
Weighing Dishes, 5 1/2*5 1/2*1	VWR-Canlab	12577-049	500	117.60
Fisher brand scoopula spatula 6"	Fisher	14-357	6 pack	8.50
Kimberly-Clark Sterilization Wrap 45*45cm	CHUL store	02-0051513	480	118.99
Pique disposable 42.5 x 60 cm	CHUL store	02-0031513	100	10.61
Strate-line autoclavable tape 24mm	CHUL store	02-0051506	1	
Printed orange high-density, high molecular polyethylene bags	Fisher	01-815A		49.51
MultiGuard Barrier Pipet Tips, Sterile (0,1-10 µl)	VWR Canlab	53550-446		206.21
200 µl tip (nuclease free)	VWR Canlab	53508-783	96/box, pk of 10	69.26
1000 µl tip (nuclease free)	VWR-Canlab	23470-004	100/box, pk of 10	51.14
200 µl tip	VWR Canlab	CA22234-320	1000/bag	123.12
1000 µl tip	VWR-Canlab	CA23470-002	1000/bag	14.34
Gilson Diamond Tip D5000	MANDEL	GF-F161370		159.95
Gilson Diamond Tip D10ml	MANDEL	GF-F161230		119.95
BD Falcon™ Disposable Polyetyrene Serological Pipet (2ml)	VWR-Canlab	CA32314-010	Case of 1000	287.15
BD Falcon™ Disposable Polyetyrene Serological Pipet (5ml)	VWR-Canlab	CA32314-008	Case of 200	79.12
Falcon 10 ml pipette	VWR-Canlab	CA32314-006	200	79.12
Falcon 50 ml pipette	VWR-Canlab	CA32314-000	100	289.43
Pasteur pipette	CHUL store	10-0011501	250	10.08
Cotton-plugged Borosilicate Glass pipets	Fisher	13-678-8B		82.90
0.5 ml Microtube, Clear	VWR-Canlab	22234-378	1000	20.28
1.5 ml Microtube, Clear	VWR-Canlab	22234-382	500	10.93
2.0 ml Screw cap tubes, Natural	VWR-Canlab	10011-598	500	214.62
Falcon 2059 tube (15 ml)	VWR-Canlab	CA60819-761		156.19
15 ml Petri Popylene conical tube with Styrofoam rack	Sarstedt	62.554.002	500	95.00

15 ml Petri Popylene conical tube	Sarstedt	62.554.205	500	85.00
Falcon Blue max 50 ml Petri Popylene conical tube	CHUL store	01-0401538	25	7.96
50 ml Petri Popylene conical tube with Styrofoam rack	Sarstedt	62.547.004	200	44.00
50 ml Petri Popylene conical tube	Sarstedt	62.547.205	250	48.00
Syringe 60 cc	CHUL store	02-0041087	1	0.3480
Filter 0.22 um for syringe	Sarstedt	83.1826.001	50	75.00
Filter 0.45 um for syringe	Sarstedt	83.1826	50	75.00
Filter 0.22 um, 500 ml	Sarstedt	83.1823.001	12	118.00
Filter 0.45 um, 500 ml	Sarstedt	83.1823.100	12	86.00
BD Falcon™ Cell Culture Plates (12 well)	VWR-Canlab	CA62406-165	50	109.20
Eppendorf Conbitip Plus Tip, Biopur (0.1ml)	Fisher	21-381-400	100	122.50
Eppendorf Conbitip Plus Tip, Biopur (0.5ml)	Fisher	21-381-402	100	122.50
Eppendorf Conbitip Plus Tip, Biopur (1.0ml)	Fisher	21-381-403	100	122.50
Eppendorf Conbitip Plus Tip, Biopur (2.5ml)	Fisher	21-381-404	100	122.50
Eppendorf Conbitip Plus Tip, Biopur (5.0ml)	Fisher	21-381-405	100	122.50
Eppendorf Conbitip Plus Tip, Biopur (10ml)	Fisher	21-381-406	100	122.50
Eppendorf Conbitip Plus Tip, Biopur (25ml)	Fisher	21-381-118	100	140.00
Adaptor for 25 ml Eppendorf Combipip Plus Tips Biopur	Fisher	21-381-308	7	95.00
Eppendorf Conbitip Plus Tip, Biopur (50ml)	Fisher	21-381-407	100	140.00
Adaptors for 50ml Eppendorf Conbitip Plus Tip	Fisher	21-381-408	7	95.00
Tube for clamp				
BD Falcon™ Cell Culture Flask (Vent) T75	VWR-Canlab	CABD353136	60	156.90
Cover Glass (rectangular)	Fisher	02-671-35	12	42.59
Syringe 10 cc	CHUL store	02-0041023	100	0.090*100
Needle 22G 1/2	CHUL store	02-0041043	100	0.23*100
Nunc cryo tube vials 1.0 ml	VWR-Canlab	CA366656	500	313.48
Tube RTU verre borosilicate 12*75	CHUL store (Fisher)	10-0011503 14-961-26	1000	21.30
Sterile PP tubes with snap caps	Fisher	14-956-1D	500	111.71

Printer Paper for Smartspec 3000, 5 pack	BioRad	170-2506		47.00
Cell lifter	VWR-Canlab	29442-200	100	106.74
Semimicro Disposable Polystyrene Cuvette	BioRad	223-9955	1.5 ml, 100	17.00
Razor Blades, Pack of 100	VWR-Canlab	55411-050		11.37
Jewelers Microforceps	Fisher	08-953F		38.05
Pinning Forceps	Fisher	10-270		7.16
Microspatula with one tapered end, one rounded end	Fisher	21-401-10	Pack of 12	40.94
Blot Absorbent Filter Paper	BioRad	170-3932	Thick, 7.5x10 cm, 50 sheet	56.00
Immun-Blot PVDF Membrane	BioRad	162-0177	26cm x 3.3m roll	336.00
Printing paper for Sonny Digital Graphic printer UP-D895, 4 roll/box	Canbera Packard Canada			160.00
Kodak BioMax Light-1 film	PerkinElmer	1788207	8x10 in	370.00
Autoradiography Pen	VWR-Canlab	52878-169	High Energy Level	62.50

List of reagents :

Item	Step	Company	Cat#	Dimension	CAN\$
Aseptone 5	0.0	Fiher	15466376	5 L	
Ethanol	0.0	CHUL			
Lysol	0.0	Techni Science	74389-97		63,00
Bovine serum albumin (FBS)	2.1	Hyclone			
Cell banker	2.1	BLC-1	100 ml		
DMEM, High Glucose, Pyruvate	2.1	Invitrogen	11995-065	3 x 500 ml	31.44
DMSO	2.1	Sigma	D2650-5X5ML	5 X 5 ml	169.50
PBS (x10 Ph 7.4)	2.1	Invitrogen	10010-002	5 L	691.13
100X Penicillin-Streptomycin solution (0.9 Nacl)	2.1	Sigma	P0781-100ML	100 ml	
Trypsin-EDTA solution, 10X	2.1	Sigma	T4174	100 ml	31.20
Vim	2.3				
Glass-Klenz	2.4	STERIS	1114-08	3.78 l	
Liquid N ₂	2.5	CHUL			
Venor®GeM Mycoplasma Detection Kit	2.6	Sigma	MP0025	1 kit sufficient for 25 tests	281.50
Trypan Blue solution (0.4%)	2.7	Sigma	T8154	100 ml	20.60
Horse serum, New Zealand Origin	3.1	Invitrogen	16050-122	500 ml	57.00
Hematoxylin solution according to Delafield for microscopy	4.2	Sigma	03971-250ML	250 ml	71.90
Glycerol	4.2	Sigma	G9012-1L	1 L	129.50

Methanol	4.2	Fisher	A452-1	1 L	96,85
			A452-4	4 L	152,35
Mouse SPARC MAb (Clone 124413), Rat IgG2B	5.1	R and D systems	MAB942	500 µg	315.00
Recombinant SPARC protein	5.1	R and D Systems	942-SP-050	50 µg	315.00
CAPS	5.2	Sigma	C2632	100g	77,50
COL1A2 (M-80) Antibody, 200 ug/mL	5.2	Santa Cruz	sc-28654	200 ug/mL	279.00
Mitochondria Isolation Kid for Cultured Cells	5.2	Abcam	ab110170		185.00
MitoProfile Total OXPHOS, 300 ug	5.2	Abcam	ab110413	300 ug	575.00
Mouse TrueBlot® Western Blot Kit, 50 uL	5.2	eBioscience	88-8887-31	50 ul	235.00
Protease inhibitor cocktail	5.2	Sigma	P8340-1ML	1 L	84.20
Protein assay kit II	5.2	BioRad	500-0002		129.00
Ready Gel Tris-HCl Gel (10-20% Linear Gradient, 4% Stacking)	5.2	BioRad			14.00
RIPA buffer	5.2	Sigma	R0278-50ML	50 ml	55.40
SeeBlue® Plus2 Pre-Stained Standard,	5.2	Invitrogen	LC5925	500 mL	
Stealth™ RNAi Negative Control Kit	6.4	Invitrogen	12935-100		700.00

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